Ryan Adidaru

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Computer Engineering student at Universitas Indonesia with strong data analytics, statistical modeling, and research skills. Experienced in interpreting large datasets, building predictive pipelines, and translating data insights into actionable product and business recommendations.

Education

Universitas Indonesia — Jakarta, Indonesia
Bachelor of Engineering in Computer Engineering — *Expected 2027*

Research Experience

Research Assistant — SYNAPSE AI Lab — Jan. 2025 – Present
Universitas Indonesia — *Jakarta, Indonesia*

- Conducting applied research in remote sensing and image analytics under SYNAPSE-AI Lab, extracting insights from multi-spectral satellite imagery using Python, Google Earth Engine, and statistical analysis.
- Co-authored an IEEE-format research paper on deep learning-based geospatial analysis, presenting findings and evaluation methodology to academic reviewers.
- Built scalable data ingestion and preprocessing pipelines for large-scale geospatial datasets, ensuring data quality and reproducibility across experiments.

Professional Experience

Data & Systems Optimization Engineering Intern — Jan. 2026 – Present
PT. Kalbe Farma, Tbk — *Hybrid, Jakarta*

- Built and maintained FastAPI backend services and PostgreSQL/Supabase data pipelines for supply chain optimization, processing large production datasets across MPS and MRP workflows.
- Developed optimization endpoints (MILP with PuLP/CBC/HiGHS) that translate business constraints — shift scheduling, machine availability, campaign limits — into data-driven planning decisions for inventory, production, and material requirements.
- Queried and transformed large SQL datasets to support analytics dashboards and reporting for stakeholder decision-making.
- Collaborated with cross-functional SWE and data teams to define API contracts, database schemas, and RPC-based data access patterns, improving integration reliability across services.
- Designed containerized deployment workflows (Docker) for reproducible development and staging environments.

Machine Learning Engineer Intern — Aug. 2025 – Oct. 2025
Caprae Capital Partners — *Remote — Glendale, California, United States*

- Developed and deployed an ML-based scoring model as a production microservice (Python/Flask), interpreting business requirements to define model features and evaluation criteria.
- Conducted functionality and performance benchmarking, translating analytical results into clear recommendations for product stakeholders.
- Built automated data pipelines for model retrieval and serving, ensuring consistent and up-to-date deployment across environments.

AI Lead — Autonomous Marine Vehicle, Universitas Indonesia — Jul. 2025 – Present
Universitas Indonesia — *Jakarta, Indonesia*

- Led a cross-functional AI team through daily standups and weekly sprints, coordinating perception, navigation, and control to deliver an autonomous system that won 2nd Runner-Up nationally.
- Designed evaluation and benchmarking frameworks for object detection models (YOLOv8/v11/v12), analyzing model performance metrics to guide deployment decisions.
- Authored technical documentation and presented results to supervisors and team members, communicating complex analytical findings in clear, actionable terms.

AI Staff — Autonomous Marine Vehicle, Universitas Indonesia — Mar. 2024 – Jul. 2025
Universitas Indonesia — *Jakarta, Indonesia*

- Trained and benchmarked multiple computer vision models on custom datasets, systematically analyzing precision, recall, and mAP to identify trends and select optimal configurations.
- Engineered data preprocessing and image segmentation pipelines using OpenCV and Python, improving detection accuracy from 70% to 85%.

Honors & Awards

- 2nd Runner-Up, Autonomous Surface Vessel Division — Indonesian National Ship Competition (KKI) 2024, Ministry of Education, Culture, Research, and Technology.

Projects

TanyaJawab | *FastAPI, React.js, Qdrant, PostgreSQL, Redis, Gemini API* May 2025

- Built an AI-powered learning platform using Retrieval-Augmented Generation (RAG), integrating PostgreSQL for structured data, Qdrant for semantic search, and containerized deployment.

NASA Space Apps Challenge 2024 | *Python, XGBoost, Pandas, Scikit-learn, Matplotlib* Sep. 2024

- Developed an end-to-end predictive analytics pipeline using NASA climate datasets (rainfall, soil moisture, solar radiation, temperature) and local market price data.
- Engineered geospatial and temporal features; trained XGBoost regression models achieving competitive RMSE for environmental and price forecasting.
- Visualized trends and model performance with Matplotlib and Seaborn; collaborated in a 4-person team to deliver a working prototype in 48 hours.

RoR (RewrittenOnRust) | *Rust, LLMs, Qdrant, Ollama* Mar. 2025

- Open-source project implementing RAG with Qdrant VectorDB for semantic data retrieval and Ollama as LLM/embedding backend.

yolo.cpp | *C/C++, CUDA, Neural Networks* Jul. 2024

- Built a YOLO-style CNN from scratch in C++ with CUDA GPU acceleration, implementing forward/backward passes without ML frameworks — demonstrating deep understanding of model internals.

Certifications

IBM Certified Applied AI Professional | Google Advanced Data Analytics | Google Foundations of Data Science | Google Advanced Data Analytics Capstone

Technical Skills

Languages: Python, SQL, C/C++, JavaScript, Rust
Data & Analytics: pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Google Earth Engine, Jupyter
Databases & Backend: PostgreSQL, Supabase, FastAPI, Flask, Redis, Qdrant, Docker
ML & CV: PyTorch, YOLO, OpenCV, Roboflow, model evaluation & benchmarking
Tools: Git, Linux/Ubuntu, neovim, Google Sheets, Microsoft Excel